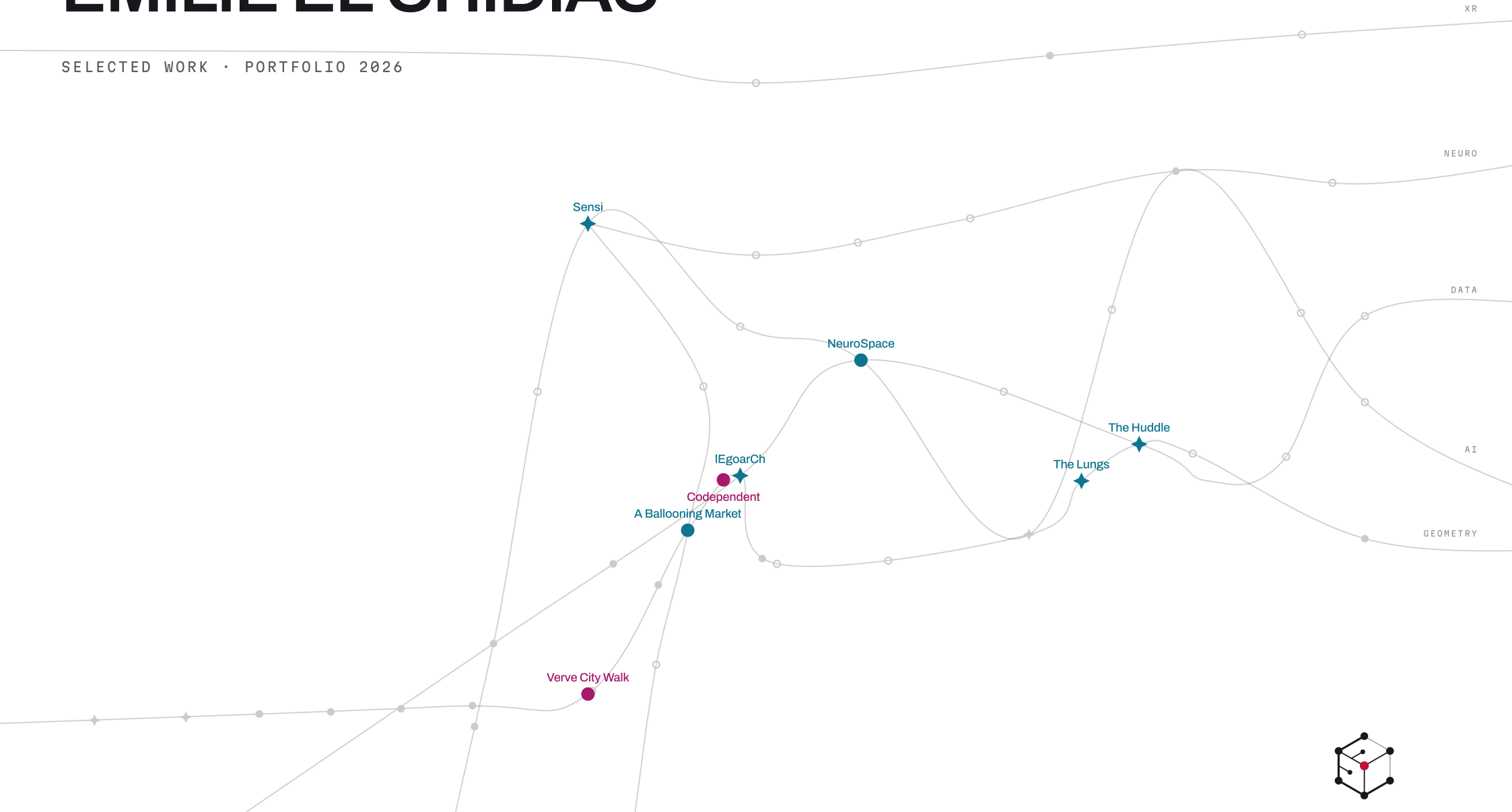


EMILIE EL CHIDIAC

SELECTED WORK • PORTFOLIO 2026



emiliechidiac.com

I work with design, technology and minds.

WHO

I will not pretend I can fill a blank notebook from nothing. What I am good at is judging what the machine hands back and steering it. Someday that might be for a cursor instead of a person. Dystopian or utopian?

We build for the eye sometimes, which helps. Mostly we build for function, which is counterintuitive, isn't it? We should be building for us to function at our best, not just the building.

So I build the instruments: Sensi scores a floor plan across six senses, NeuroSpace estimates how a room reads to the brain and shows its weights so you can disagree with them.

It was never about being the most interesting person in the room. But the most interested one: what is x doing to y ?

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Sensi

When you fix one sense in a floor plan, what happens to the other five?

TYPE computation & research
AWARD macad awards 2026 · design copilots · winner
WHERE iaac · macad studio
WHO team of 4
MY PART project lead; built agent, evals, react ui, deploy
STACK python · langgraph · mcp · fastapi · react
LINKS [live app](#) · [github](#) · [blog](#)

WHAT

Every tool in the stack could tell us how a building performs. None of them would say how a room feels. Sensi closes that gap: a copilot that reads a floor plan and scores comfort across six senses (thermal, visual, acoustic, spatial, olfactory, tactile), calibrated to one person at a time, not an average. Project lead in a team of four (Charles Abi Chahine, Lakzhmy Mari Zaro, María Sánchez Domínguez and me): we framed the question together.

WHY

Comfort is usually the thing that shows up, or does not, after the design is done. Sensi makes it a layer you can interrogate while the plan is still soft, because you do not walk into a room and average your experience: the thing that is wrong is the thing you notice.

HOW

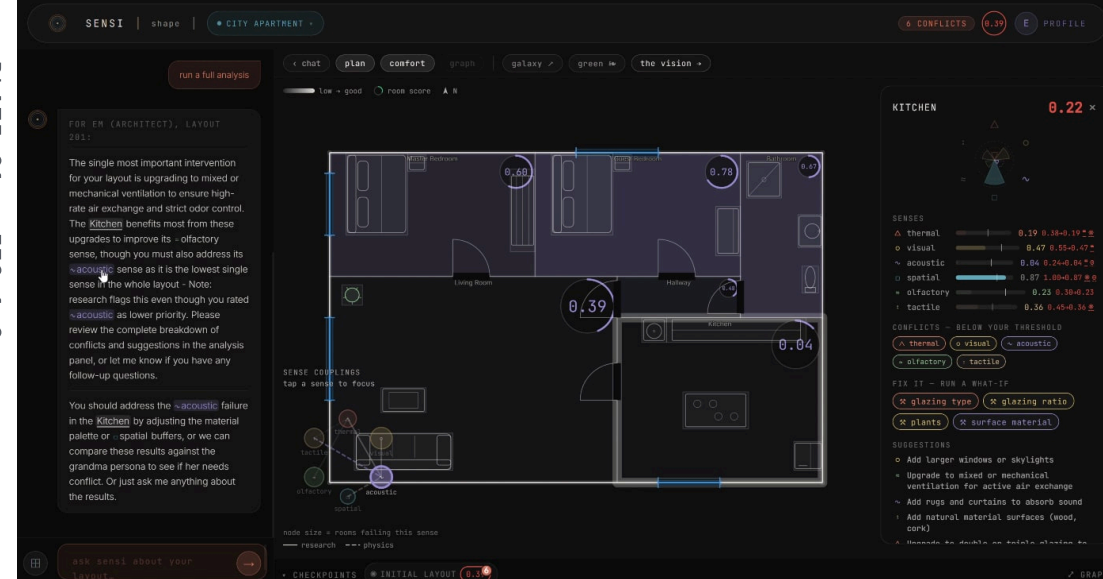
Calibrate it to one person at onboarding: thermal grudges, noise tolerance.

Route each request, one LLM call per turn, through a LangGraph graph: analyze, edit, preview, audit. Two evals ride inside: a critic on every suggestion, an evaluator on every reply.

Ripple every change, through a coupling matrix, into the neighboring senses, so a fix that quietly breaks another score gets flagged, not hidden. *n.b. the six scores argue like a family. the coupling matrix is the dinner table.*

Preview edits before they commit, then let a vision model redraw the room, structure intact, and hand over the report.

PLATE 01 · FIG 1.0



WHAT CAME OF IT

Two LLM providers scored the same three scenes, one arranged to fail, and mostly agreed. Easy to call that validation. We wrote agreement is not truth into the notes instead, and kept every disagreement as data. It hardened into evals. It estimates, it does not measure, and every coupling in the code is labeled verified or inferred, so you can see which links rest on research and which on our reasoning.

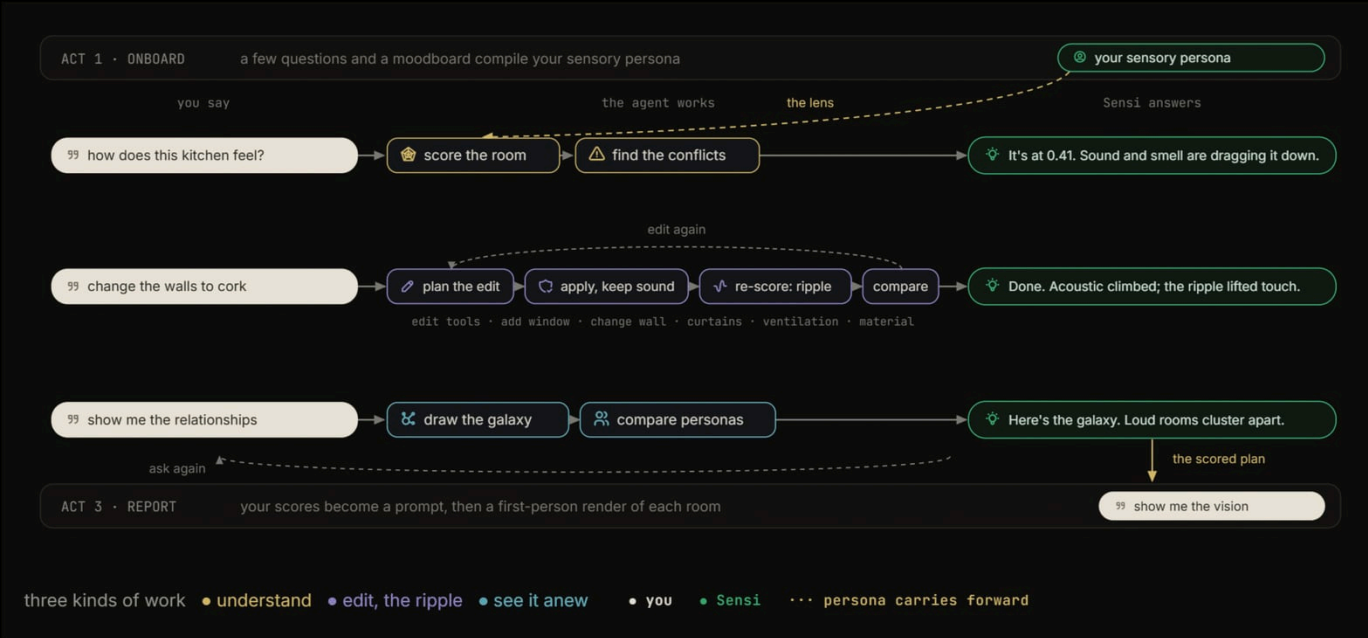


FIG 1.1 · Act 2 of the pipeline as a flowchart: score, edit, ripple and galaxy loops around the plan while it is still soft

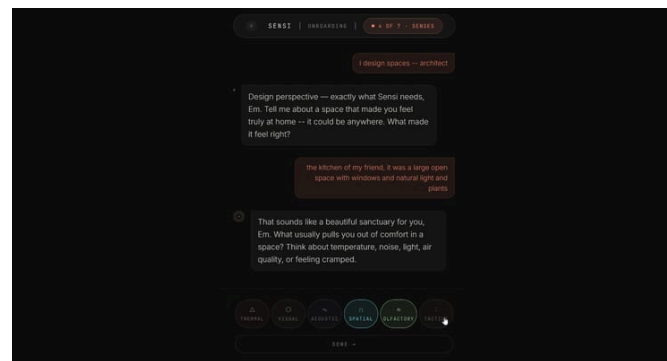


FIG 1.2 · Sensi's onboarding chat calibrating the comfort copilot to one person: their noise tolerance, their thermal grudges

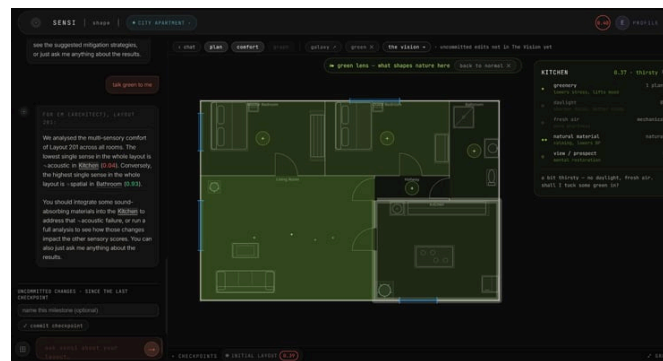


FIG 1.3 · The green lens over a city apartment plan: what shapes nature here, the kitchen flagged thirsty for daylight and plants

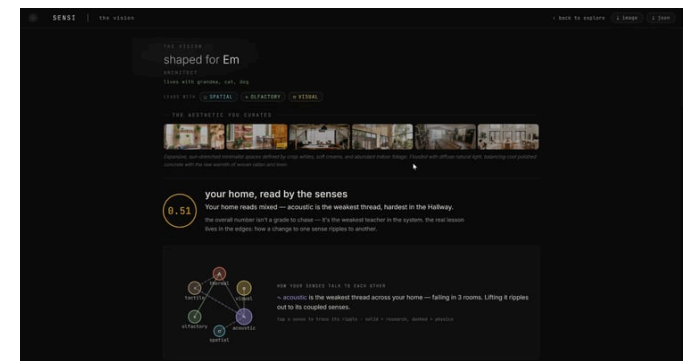


FIG 1.4 · The vision report Sensi hands over: comfort scores shaped for one person, with the edits the layout absorbed

lEgoarCh

How can AI turn a text prompt into a LEGO set that actually snaps together?

TYPE computation & research
AWARD jury award
WHERE iaac · macad generative ai
WHO with charles abi chahine
MY PART the brick solver and the whole interface
STACK flux.2 klein · trellis-2 · lora · ldraw
LINKS [github](#) · [blog](#)

WHAT

A render is a promise, not a product: you cannot snap a JPEG together on your living-room floor. lEgoarCh takes a text prompt and returns a LEGO Architecture set that is digitally verified buildable: AI imagines it, deterministic code makes it snap together, brick by brick, out of real catalog parts. Built with Charles Abi Chahine. I stitched the stages into one flow, prompt in, set out.

WHY

The inspired gesture is cheap now; anyone can generate a thousand renders before lunch. The interesting problem moved to verification: what does it take to turn a generated image into something that provably holds together?

HOW

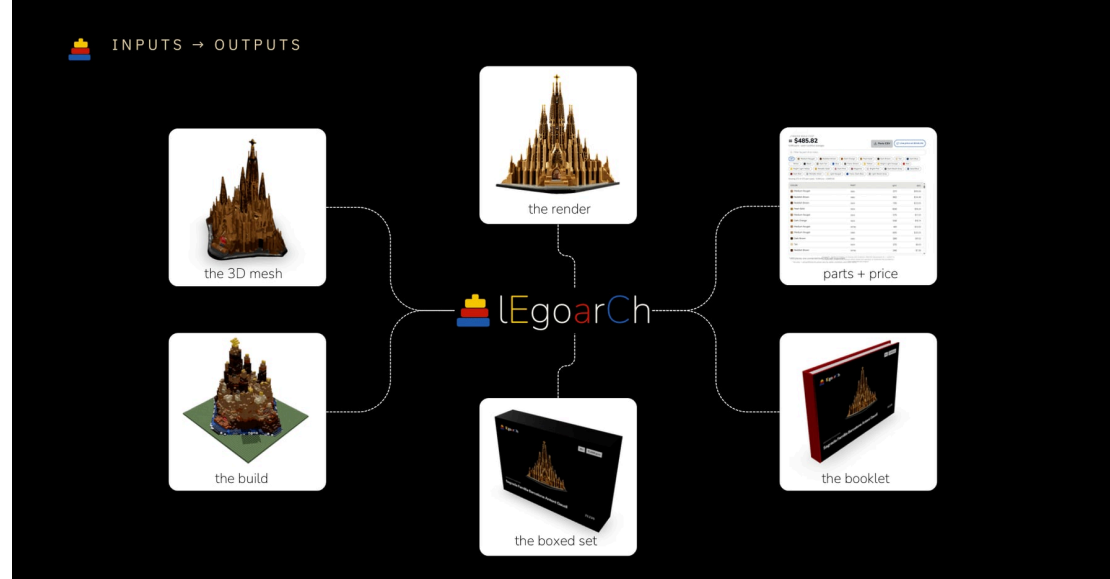
Fine-tune a LoRA on a 40-image dataset to teach FLUX the LEGO Architecture look; the prompt becomes a styled render.

Lift the render into a 3D mesh with TRELLIS, and voxelize the mesh into brick space.

Place real catalog bricks into the voxels with an optimizer, enforcing connectivity, support and perceptual color accuracy.

Export the set as LDraw, the format the brick world already speaks.

PLATE 02 · FIG 2.0



WHAT CAME OF IT

The most instructive moment was a failure: an intermediate model came back connected and supported, yet not legible as architecture. Structurally sound and visually wrong is still wrong, so legibility joined the constraints. The final pipeline was benchmarked on three buildings.

PLATE 02

lEgoarch

IAAC · MACAD GENERATIVE AI · WITH
CHARLES ABI CHAHINE

FLUX.2 KLEIN · TRELLIS-2 · LORA ·
LDRAW

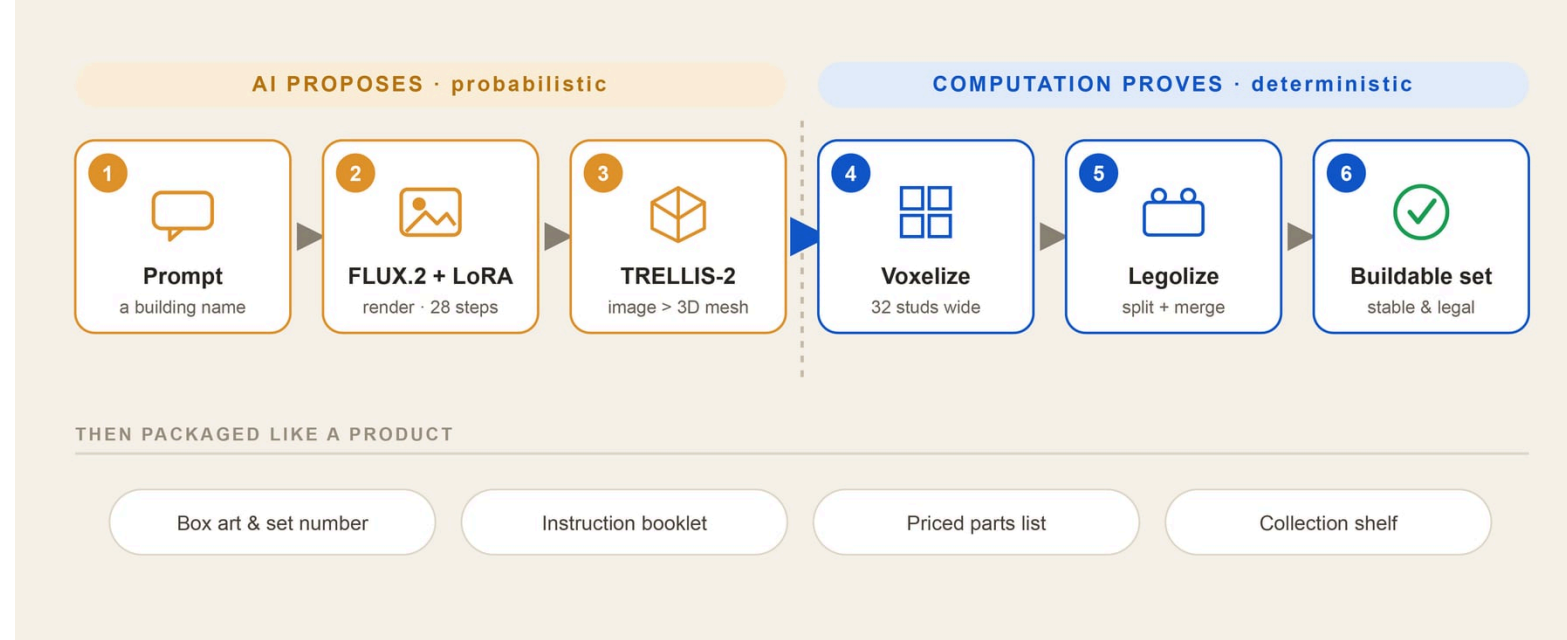


FIG 2.1 · The lEgoarch GenAI pipeline: FLUX and TRELLIS propose, deterministic code verifies the LEGO set as digitally buildable

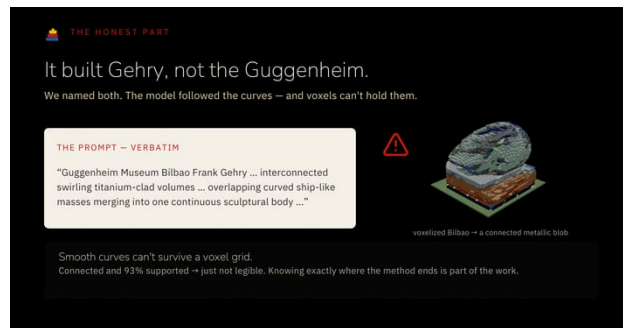


FIG 2.2 · The failure that taught the constraint: the Guggenheim came back connected and supported, and still not legible as architecture

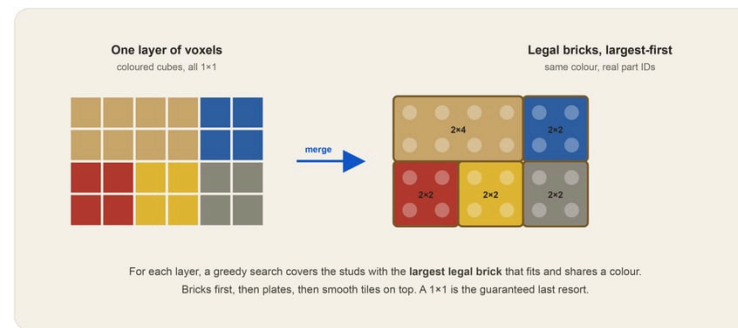


FIG 2.3 · The split and merge pass packing the voxel grid into the largest legal bricks, fewer parts holding the same shape



FIG 2.4 · Real parts and real molds: 44 BrickLink parts in 48 colors, the catalog the optimizer is allowed to build from

NeuroSpace

Can you visualize the parameters that affect how a room makes you feel?

TYPE computation & research
WHERE iaac · macad
WHO solo
STACK vue 3 · three.js · force density
LINKS [live app](#) · [github](#) · [blog](#)

WHAT

You are sitting in a room right now, and its defaults are quietly working on you: the ceiling height nudging your cortisol, the daylight setting your circadian clock. NeuroSpace makes that invisible layer legible: change one thing, and a membrane room form-finds itself live while a score answers back. The first version ran its geometry through Grasshopper, on a server that has since died. This one runs entirely in your browser.

WHY

This is the thesis I keep circling: BIM, reframed from Building Information Modeling to **Behavior Information Modeling (BeIM)**. The information that matters is not just what a building is made of; it is what the building is doing to the person inside it.

HOW

Describe the room as parameters, not geometry: ceiling, walls, openings, organic form, plants, each a slider.

Form-find it as one tensioned membrane by force density, the method written for Frei Otto's Munich Olympic roof: each drag re-solves it in milliseconds.

Score it in the browser against a control room: a transparent weighted sum you can split the screen with, or flip to by holding F.

Write every rule in code with Claude Code as a pair: I set the rule, then check it against its source.

PLATE 03 · FIG 3.0



WHAT CAME OF IT

It shipped live, then the server under it died. I rebuilt it instead of letting it go, partly for Dr. Cleo Valentine's note, the frame I kept: treat every score as a hypothesis. So the lab now tests one at a time, against a control. Partly as an excuse to form-find without Grasshopper. The weights sit in the public repo, to argue with. *n.b. a score you can argue with beats a number you have to trust.* The score stays a heuristic: it never measures your body.



PLATE 03

NeuroSpace

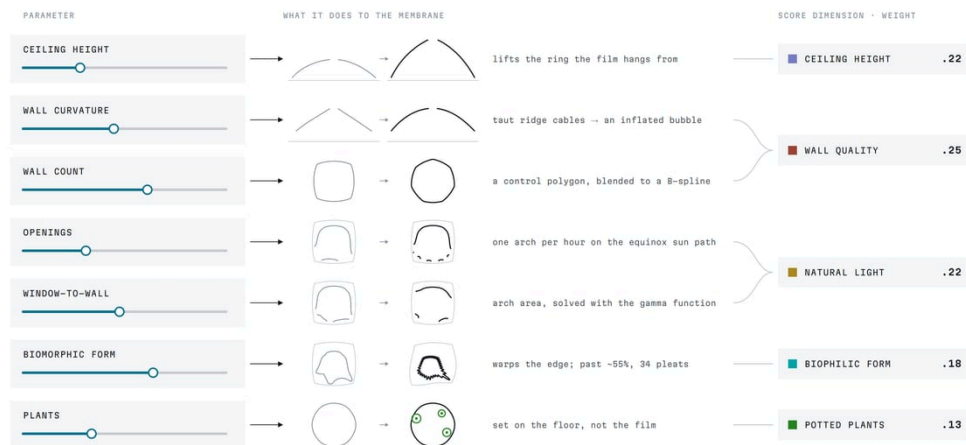
IAAC · MACAD · SOLO · LIVE APP

VUE 3 · THREE.JS · FORCE DENSITY

FIG 3.1 · The model as a matrix: each slider's weight into each dimension, readable in the app and in the public repo

THE RULES · FROM SLIDER TO SCORE

each glyph drawn by the app's own solver: before in grey, after in ink

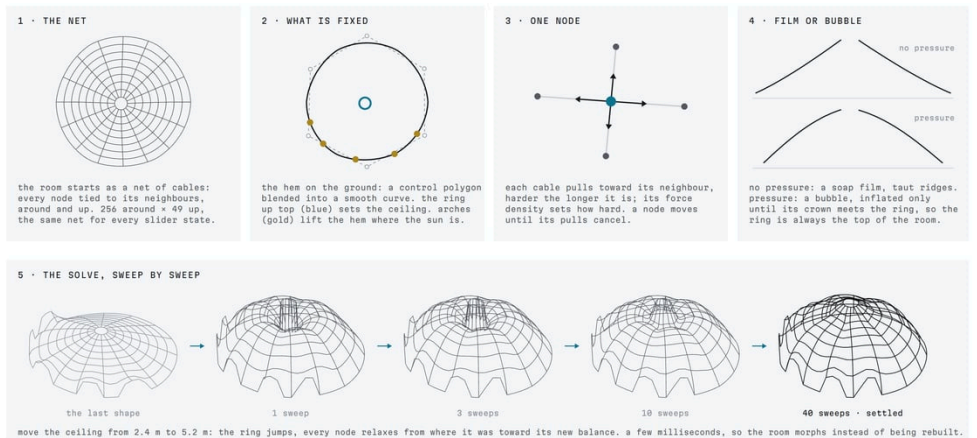


NEUROSCORE = a transparent weighted sum of the five, .22 + .25 + .22 + .18 + .13 = 1 · an estimate from published research, never a measurement

FIG 3.2 · The rules, drawn by the app's own solver: each slider, what it does to the membrane, and the score dimension it feeds

FORM-FINDING · HOW THE ROOM FINDS ITS SHAPE

the force density method, drawn from the app's own solver: the logic, not the maths



force density: Schek, 1974, written for Frei Otto's Munich Olympic roof. The first NeuroSpace got this shape from a Grasshopper definition on a server; here it is one readable file of code, src/geometry/membrane.js, running in your browser.

FIG 3.3 · How the room finds its shape: the net, what is fixed, one node's pulls cancelling, and the solve relaxing sweep by sweep

The Lungs

Can a dashboard run an architecture studio?

TYPE computation & research
AWARD studio award
WHERE iaac · macad studio
WHO data team of 3
MY PART ux, timeline, kpi map, stress-test game, guided tour
STACK vue 3 · vite · tailwind · speckle
LINKS [live app](#) · [blog](#)

WHAT

Santiago has an air problem, the kind that sits in the valley like a guest who will not leave. The Lungs is a hyperbuilding designed to filter 12 million m³ of air a year, and the data team of three (Lakzhmy Mari Zaro, María Sánchez Domínguez and me) built the live web app the whole studio ran on while designing it.

WHY

Three design teams, ten weeks, one shared organism: somebody had to build its nervous system, and that turned out to be the most architectural thing we designed.

HOW

Build the platform in Vue 3, Vite and Tailwind: KPIs, team status and one timeline per team, so the overlaps show before they collide.

Stream the 3D models in through Speckle, so performance data sits on geometry, not beside it.

Keep it running while the studio designs on top of it: not documentation after the fact, the thing the studio ran on.

Score calm: a stress game the studio actually played, half telemetry (who is really using the app), half care (how everyone is holding up).

WHAT CAME OF IT

The platform tracked three design teams across the full ten-week studio, and it is still live, with two doors: a public page where anyone can watch the tower's data, and a managerial side behind IAAC emails. Building a platform alongside the project it supports is messy, the teams' data outran the app more than once, and the mess is the lesson: the building breathes, so does the app.

PLATE 04 · FIG 4.0

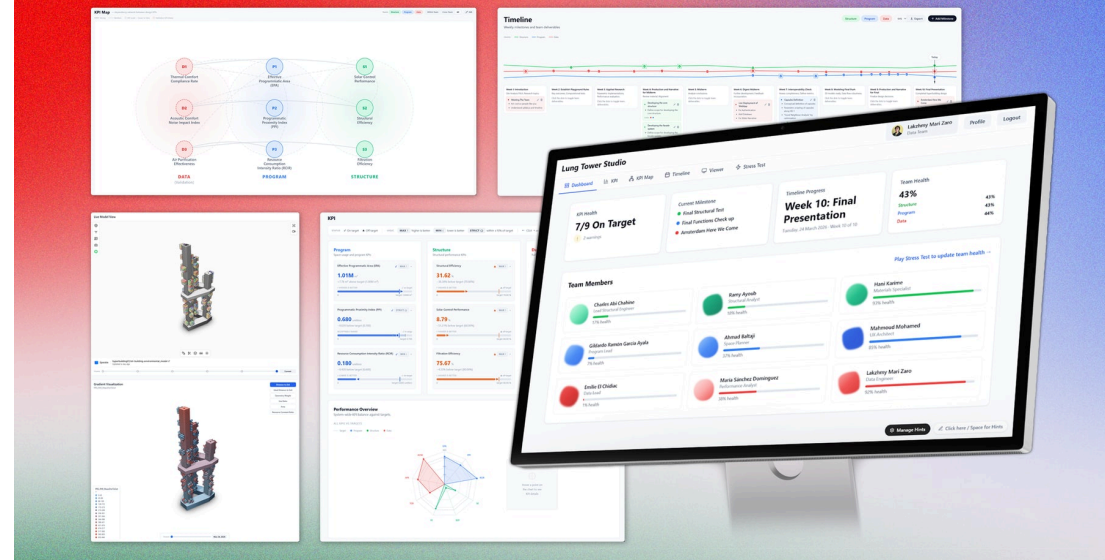


PLATE 04

The Lungs

IAAC · MACAD STUDIO · DATA TEAM
OF 3 · LIVE APP

VUE 3 · VITE · TAILWIND ·
SPECKLE



FIG 4.1 · Ten weeks as a live timeline: three team tracks whose lines converge wherever the teams overlap, and the weekly deliverables the app kept honest

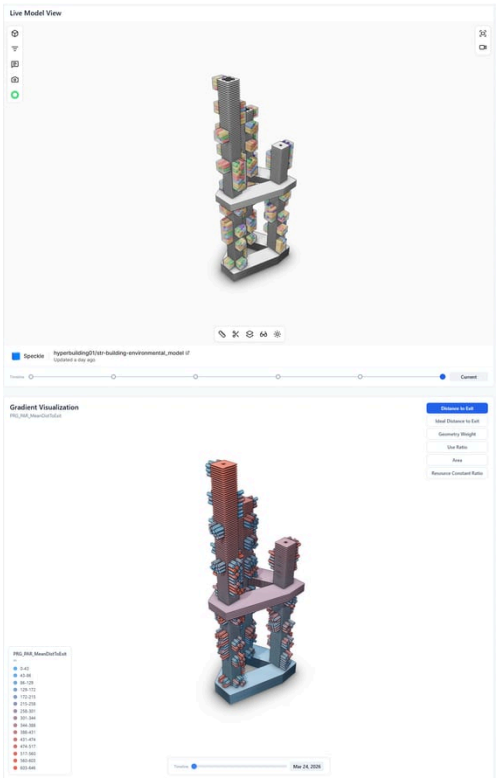


FIG 4.2 · Speckle streams the design teams' tower model into the app, so performance gradients sit on the geometry itself

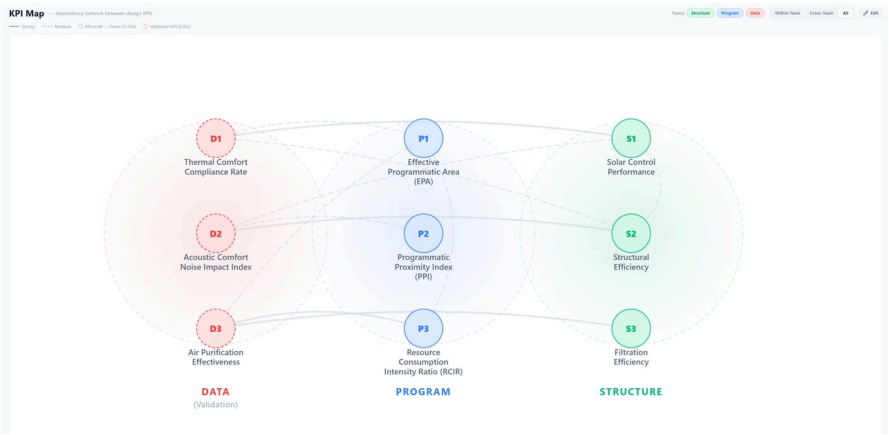


FIG 4.3 · The KPI dependency network: data, program and structure indicators tied into one system the studio steered by

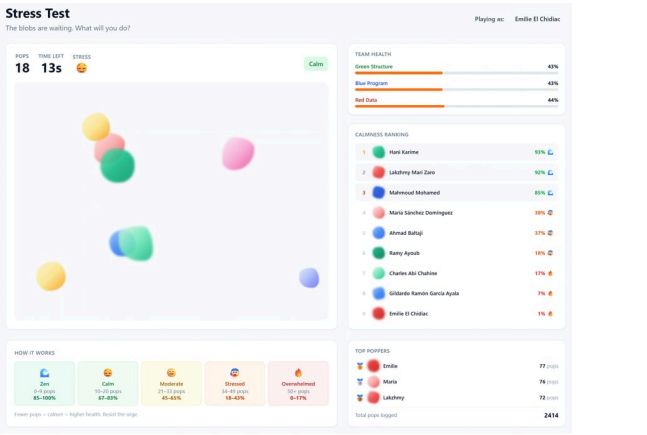


FIG 4.4 · The stress test tab, a calmness leaderboard the studio actually played; fewer pops means calmer

The Huddle

What if you built with the wind instead of bracing against it?

TYPE computation & research
AWARD studio award
WHERE iaac · macad studio
WHO team of 4
MY PART wasp growth study and panel distribution
STACK wasp · kangaroo · alpaca4d
LINKS [blog](#)

WHAT

In Punta Arenas the wind never stops, so we stopped fighting it. The Huddle is a research and education hub grown from 4×4×4 m modules aggregated along the wind itself, wrapped in an envelope of three panel types (Shields, Lenses, Gills), each answering a different face of the weather. A team of four: Charles Abi Chahine, Lakzhmy Mari Zaro, María Sánchez Domínguez and me.

WHY

Wind is the site's most abundant force, and the vernacular Kawésqar huts already knew what to do with it. Accepting the wind as a design partner, instead of a problem to brace against, is the whole project: the harshest thing on the site becomes the thing that decides where everything goes.

HOW

Grow the module cluster with WASP along the wind patterns, solar exposure and program, the way the Kawésqar huts huddled.

Settle the layout in Kangaroo; check the structural behavior of the result in Alpaca4D.

Distribute the three envelope panels with a Global Index algorithm, so the facade reads the climate back to you.

Set vertical turbines in the aggregation's own wind tunnels, harvesting the force the form was grown from.

WHAT CAME OF IT

Wind is physics, so the planning became physics too: collision and settling dictated the groupings, and no module was placed by hand. We controlled the aggregation by tuning the forces, which is the only way it did not control us.

PLATE 05 · FIG 5.0



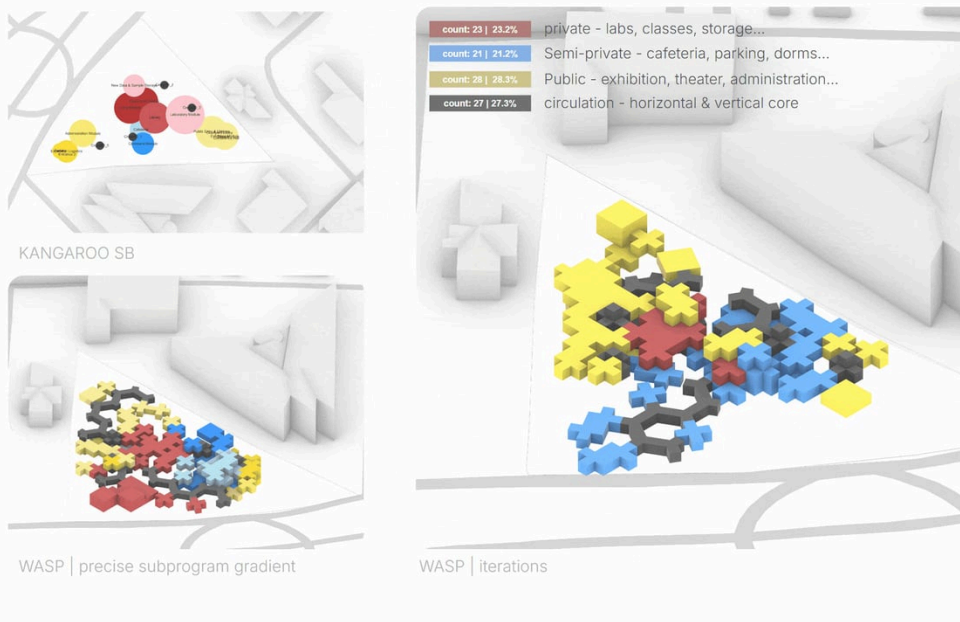


FIG 5.1 · Stills from the WASP growth study: modules aggregating along wind and program across the Punta Arenas plot

PLATE 05

The Huddle

IAAC · MACAD STUDIO · TEAM OF 4

WASP · KANGAROO · ALPACA4D

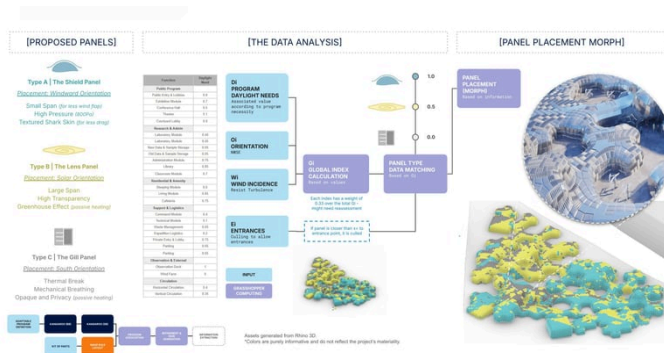


FIG 5.2 · Global Index analysis matching Shield, Lens and Gill panels to daylight, wind and program across the envelope



FIG 5.3 · Blizzard courtyard render inside The Huddle, turbines spinning between the wind-formed modules in Punta Arenas



FIG 5.4 · Street-level render of the module cluster meeting the town, the Punta Arenas sign in the foreground

A Ballooning Market

Can balloons roof a historic market without touching it? A pneumatic simulation.

TYPE computation & research
WHERE iaac · macad
WHO solo
STACK grasshopper · kangaroo · dendro · d5
LINKS [blog](#)

WHAT

A historic Cairo market receives a new roof of pressure-packed balloons that borrows the existing steel frame without modifying it. *n.b. the frame is the client, the balloons are the tenants, the solver is the lease.* Each balloon is a Kangaroo body with collision, inflation and anchor goals; the settled cluster is meshed with Dendro and lit through CMY membranes in D5.

WHY

A building like Bab al-Luq usually gets offered two futures: a museum piece that can never be touched, or a cold glass-and-steel box dropped in the middle and called modern. Pneumatic parasitism is the third option: a soft new life that hugs the old bones instead of replacing them.

HOW

Trace the frame of Bab al-Luq; mark the nodes that can host anchors.

Seed balloons in the volume; declare radii, no physics yet.

Add collision and inflation goals; anchor the cluster; let the solver settle.

Mesh the settled cluster with Dendro; render daylight through the CMY membranes in D5.

WHAT CAME OF IT

Tuning became the craft: collision too low and the balloons cuddle, too high and they panic. You tune it like a thermostat. And the run where they had no awareness of each other at all, just ghosting through one another in a chaotic, colorful mess? It stayed in the record on purpose, first in the gallery: people read honesty faster than polish.

PLATE 06 · FIG 6.0



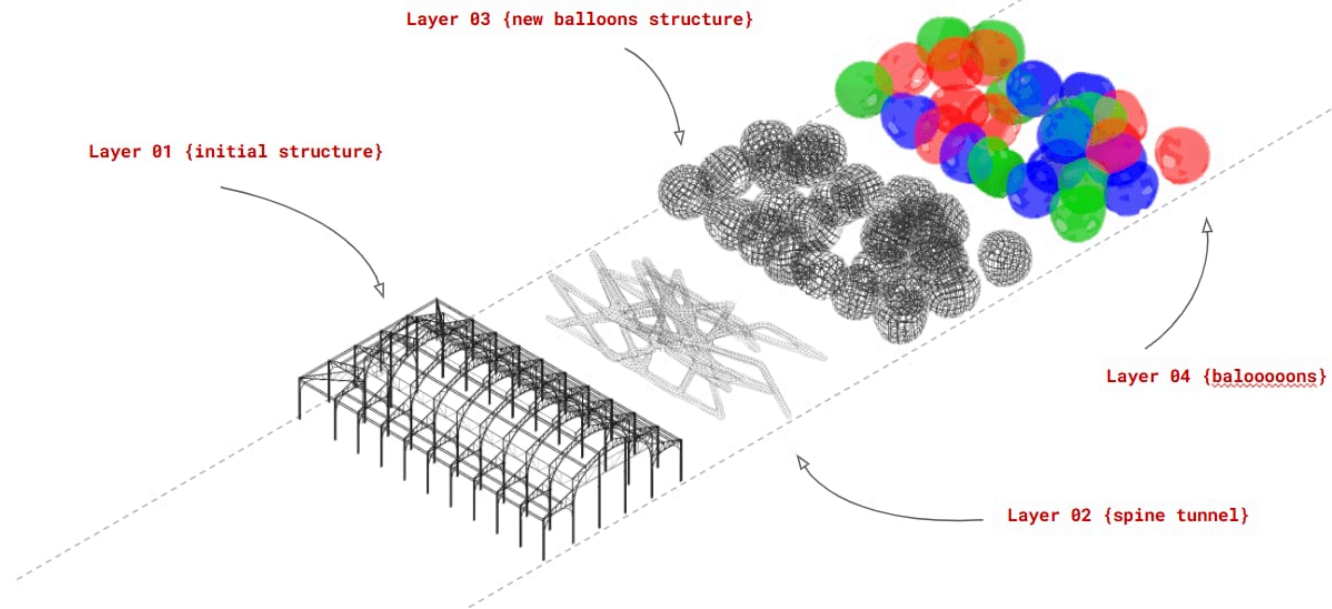


FIG 6.1 · Exploded layer diagram of the pneumatic occupation: historic steel frame, spine tunnel, and the inflated balloon roof

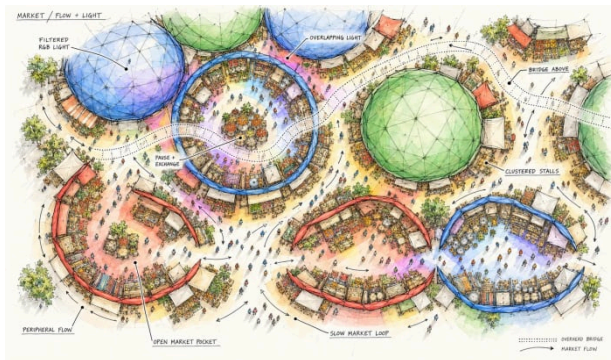


FIG 6.2 · Market, flow and light in watercolour plan: the balloons curve pockets, a slow market loop, the bridge above

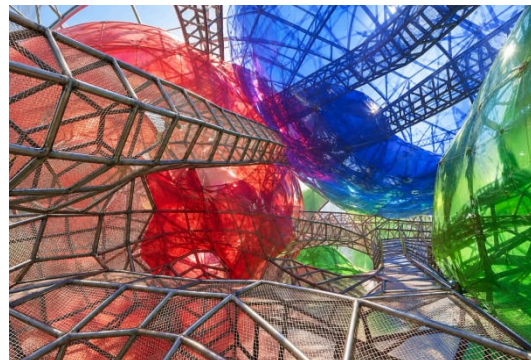


FIG 6.3 · Inside the net tunnel: the steel walkway threads between red, blue and green balloons toward the open sky

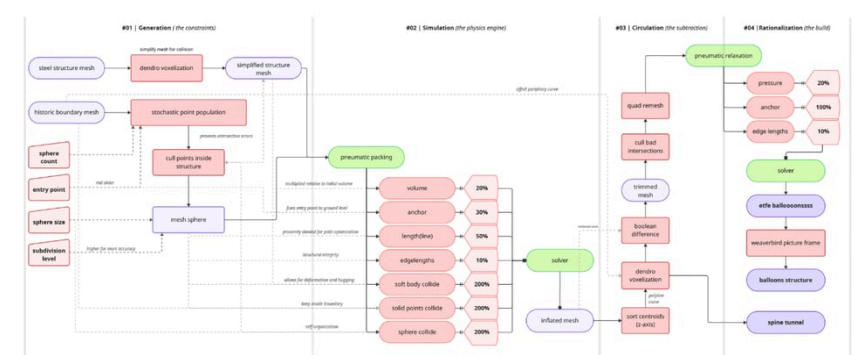


FIG 6.4 · The Grasshopper algorithm behind the inflation simulation, from constraints to Kangaroo goals to the settled cluster

Codependent

What if no piece could stand on its own?

TYPE design & practice
WHERE dynamic solution · kuwait
WHO practice team
MY PART design lead: concept, grasshopper, cnc cutting files
STACK rhino · grasshopper · cnc cutting · unreal · vr

WHAT

It begins with a simple idea: each piece needs another to stand. Two interlocking wooden panel types form a modular assembly without glue or screws between pieces. An exhibition stand on a 6 by 6 m platform: 780 mm panels, white and black, slot into a structure 2.94 m high, and the same system becomes seats at 408 mm and counters at 948 mm.

WHY

Mutual support becomes both a construction method and a place to gather. The same system creates seats, counters and open edges where people can stop, meet and move through.

HOW

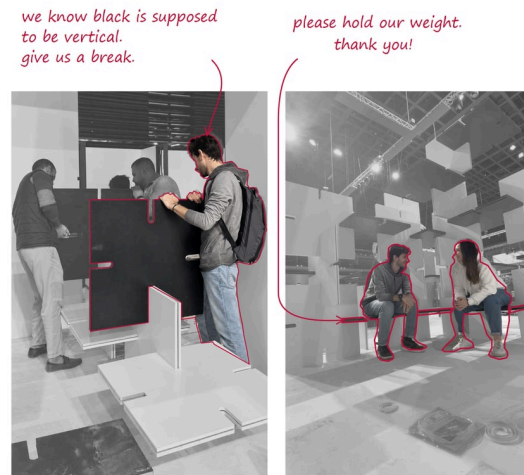
Explore how that system can grow in the Grasshopper definition, using connection rules, spatial limits and support checks.

Walk the client around the options in Unreal VR to settle the priorities before the build: how many seats, how many counters, at what height.

Narrow thirty arrangements to ten, and ten to one: option 29, 34 parts.

Turn the selected option into numbered parts and CNC cutting layouts, each part ID tied to its geometry, assembly and parts schedule.

PLATE 07 · FIG 7.0



We designed a puzzle. Then some of the pieces went missing.

A little too committed to the concept, apparently.

No spares. So we worked out new connections on site, using the pieces we had.

What could move, what could still fit, and how could we keep the seats and counters useful?

- 01 the pieces we had
- 02 new connections
- 03 still somewhere to sit + meet

WHAT CAME OF IT

Option 29 went to site as 34 panels with no spares, and some were misplaced. We went back and cut new ones, but pieces were already connected, so the layout had to be improvised around them: the booth stood with 39, still balanced. We designed a puzzle, and then had to solve it together. Afterwards the panels moved into the brand's office as small grouped pieces.



FIG 7.1 · The booth at night by the sea in Kuwait: interlocking white panels and black shelves form seats, counters and open edges

PLATE 07

Codependent

DYNAMIC SOLUTION · KUWAIT · PRACTICE TEAM

RHINO · GRASSHOPPER · CNC CUTTING · UNREAL · VR

- 01 Slot together.**
The joint is cut into the panel.
No glue or screws between pieces.
- 02 Support one another.**
Vertical pieces carry the horizontal.
Crossing slots restrain the assembly.
- 03 Make room to gather.**
The lower surface becomes a seat.
The higher one becomes a counter.

The same two pieces make the connection, the support and the furniture.

FIG 7.2 · The idea in three moves: slot together, support one another, make room to gather on a lower seat and a higher counter

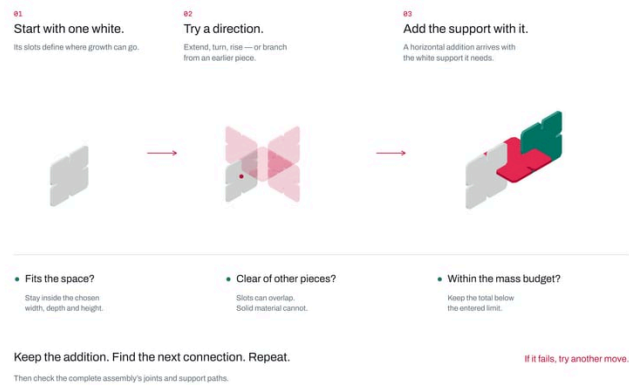


FIG 7.3 · The growth rule: start with one white, try a direction, add the support, then check space, clearance and mass



FIG 7.4 · Thirty generated arrangements laid out, turned to plan, narrowed to ten candidates and then to one: option 29

Verve City Walk

What survives when a parametric study meets a real drawing set?

TYPE design & practice
WHERE soma
WHO practice team
MY PART design architect: grasshopper facades, revit bim
STACK rhino · grasshopper · revit
LINKS [verve @ meraas](#) · [verve @ soma](#) · [district o @ soma](#) ·
[enara @ soma](#) · [saria @ beyond](#)

WHAT

Verve is a two-tower high-rise in Dubai: a shared amenities podium under both towers, balconies cut into the facade rather than hung off it. As a design architect at SOMA I ran the parametric facade studies in Rhino and Grasshopper, then carried what survived into the Revit model: floorplans, facade strategies, the interiors of a residential tower. The same desk carried District O, Enara and Saria from massing studies into their BIM sets.

WHY

These are the years when I learned that a parametric study only matters if it survives contact with a drawing set. Practice is where the elegant definition meets the door schedule, and both have to win.

HOW

Move everything into Revit at design development: landscape planning, the typical floor plates, the interior layouts of a luxury tower. This is where gross floor area starts running the project.

Build Revit families with the visualization team, and send the same model out three ways: 3D prints, renders, a VR walkthrough in Unreal.

Then bring every consultant back into the file, structure, MEP, facade, because the Revit model is the submission and it has to be true on the day it leaves.

WHAT CAME OF IT

At university nobody pays for the square meters, so nobody counts them. Then you are in practice and gross floor area is the only number in the room: every balcony, every core shifted, every planted terrace, weighed against that one figure. The proof of the coordination came the day a client put on a headset and walked a tower that did not exist yet.

PLATE 08 · FIG 8.0



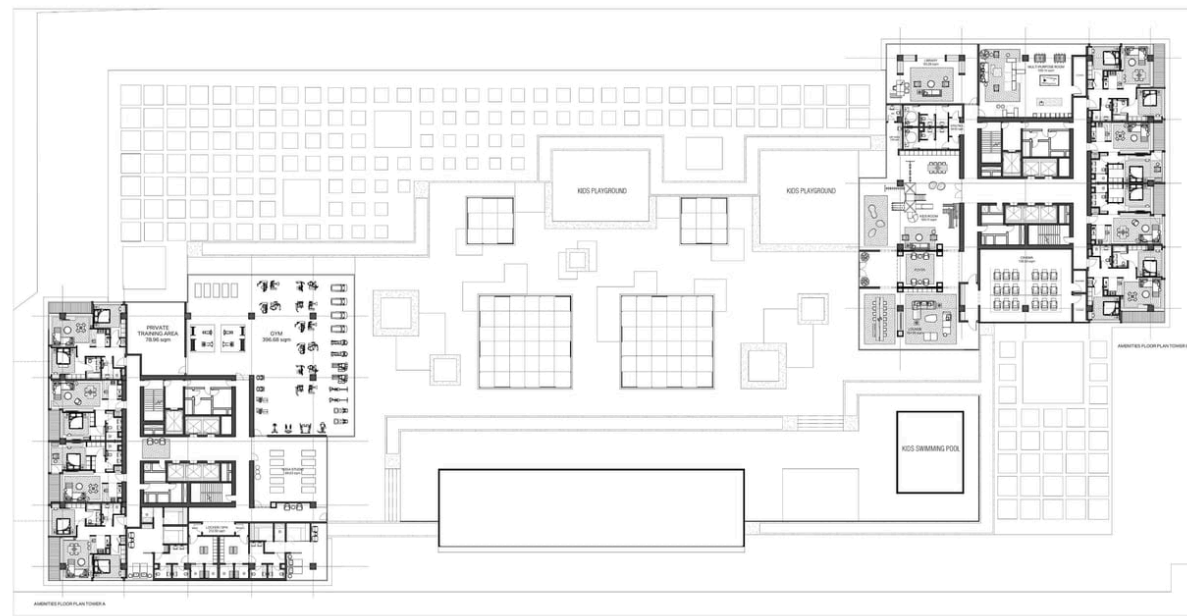


FIG 8.1 • Amenities level plan for the two Verve towers: gym, spa, yoga studio, cinema, kids' pool and playgrounds on the podium



FIG 8.2 • Screen capture of the interactive 3D unit selector orbiting the twin Verve towers, one unit footprint highlighted

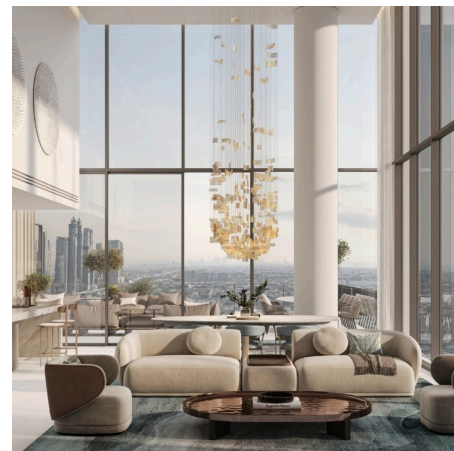
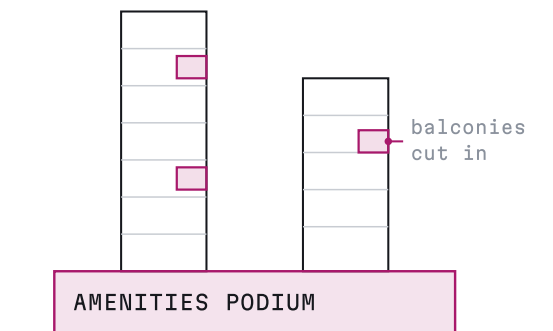


FIG 8.3 • Interior render of a double-height Verve lounge, a suspended chandelier and floor-to-ceiling glass over the skyline

TWO TOWERS, ONE PODIUM



gym • spa • yoga • cinema
kids' pool • playgrounds

FIG 8.4 • Two towers of different height over one shared amenities podium, with the balconies cut into the facade rather than hung off it

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DYNAMIC	SOMA	SELF · +	LAU · +	JEMMA · +	LAU	JEMMA	

THE LATEST THOUGHTS

○ we see in silence	sep 2026	○ where the plant is	aug 2026	○ explaining things	jul 2026
○ adjacency is not access	jun 2026	○ buildings that respond	jun 2026	○ experiences are data	may 2026
○ comfort as data	may 2026	○ when the tool scores people	apr 2026	○ latent space	apr 2026



this is what's on my mind. what's in yours?

`chidiacemilie@gmail.com` · `linkedin.com/in/EmilieElChidiac` · `github.com/hi-em`

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